Four Ways to Show a Line

equation · table · graph · words

Equation

$y = 2x + 1$

slope 2, y-intercept 1

Words

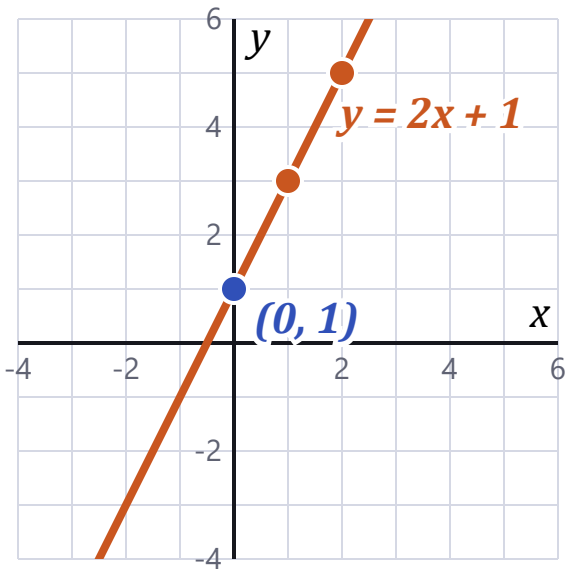
Start at 1. Add 2 each time.

Table

x	y
0	1
1	3
2	5
3	7

Each time x goes up 1, y goes up 2.

Graph



Every form shows the same two things: the **slope** (rate of change) and the ***y*-intercept** (starting value).

Find the Slope and Start

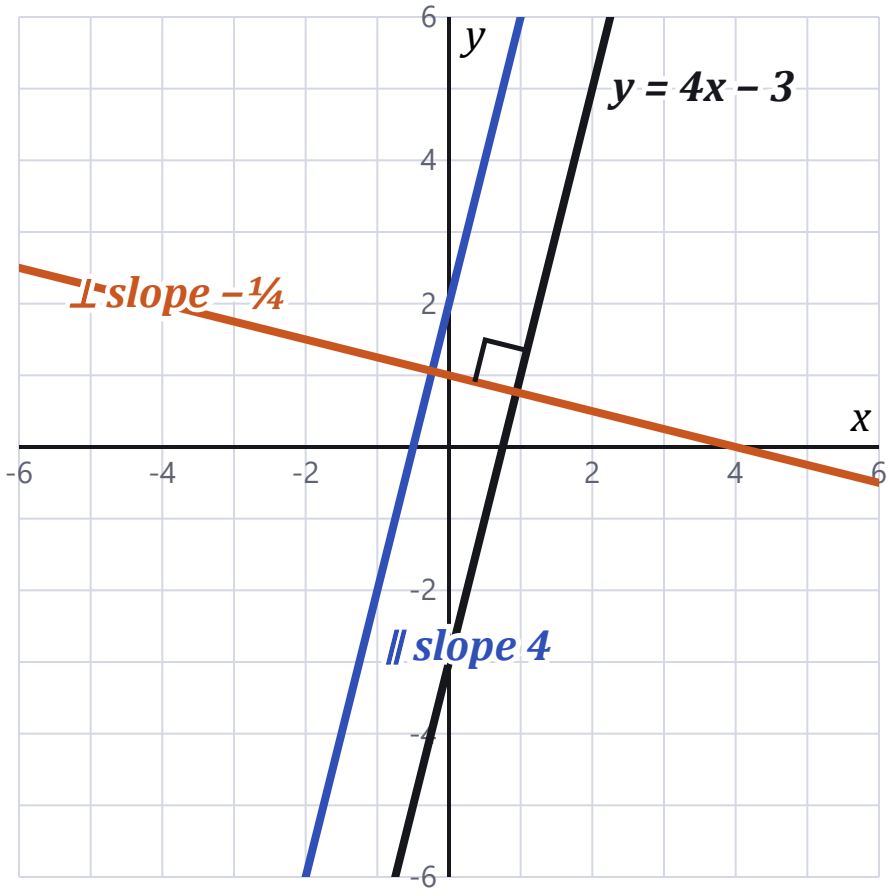
$$y = mx + b$$

Form	Slope (rate)	*y*-intercept (start)
Equation	the number times x $y = 3x + 10 \rightarrow 3$	the number added $y = 3x + 10 \rightarrow 10$
Table	change in $y \div$ change in x $(0, 3), (2, 11) \rightarrow 8 \div 2 = 4$	y when $x = 0$ $(0, 3) \rightarrow 3$
Graph	rise ÷ run between two points	where the line crosses the *y*-axis
Words	*per, each, every* $25 **each** month → 25	*starts with, fee* **starts with** $25 → 25

Slope formula: $m = (y_2 - y_1) \div (x_2 - x_1)$. Use any two points from a table or a graph.

Parallel or Perpendicular?

Compare to $y = 4x - 3$



Same slope

Parallel ∥

Product is –1

Perpendicular ⊥

$4 \cdot (-\frac{1}{4}) = -1$

Anything else

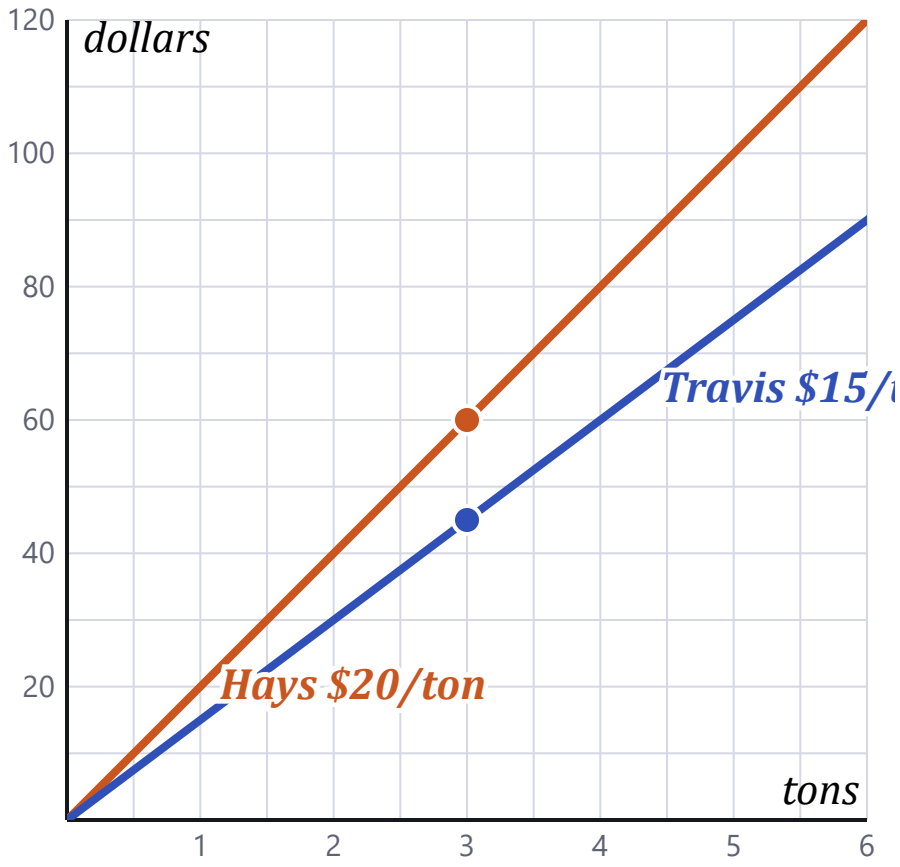
Neither

$4 \cdot \frac{1}{4} = 1$, not -1

Function	Slope	Compared to slope 4
Chiletso bakes 4 dozen cookies every hour.	4	**parallel**
Table: $(-4, 3), (0, 2)$	$-\frac{1}{4}$	**perpendicular**
$y = \frac{1}{4}x + 5$	$\frac{1}{4}$	**neither**

Which Grows Faster?

Steeper line = bigger rate



Recycling cardboard

Travis County (table)

Tons	Payout
1	$15
2	$30

$15 per ton

Hays County (graph)

through (1, 20)

$20 per ton

Compare the slopes

20 > 15, so **Hays pays more** for every ton: 3 tons → $60 vs. $45.

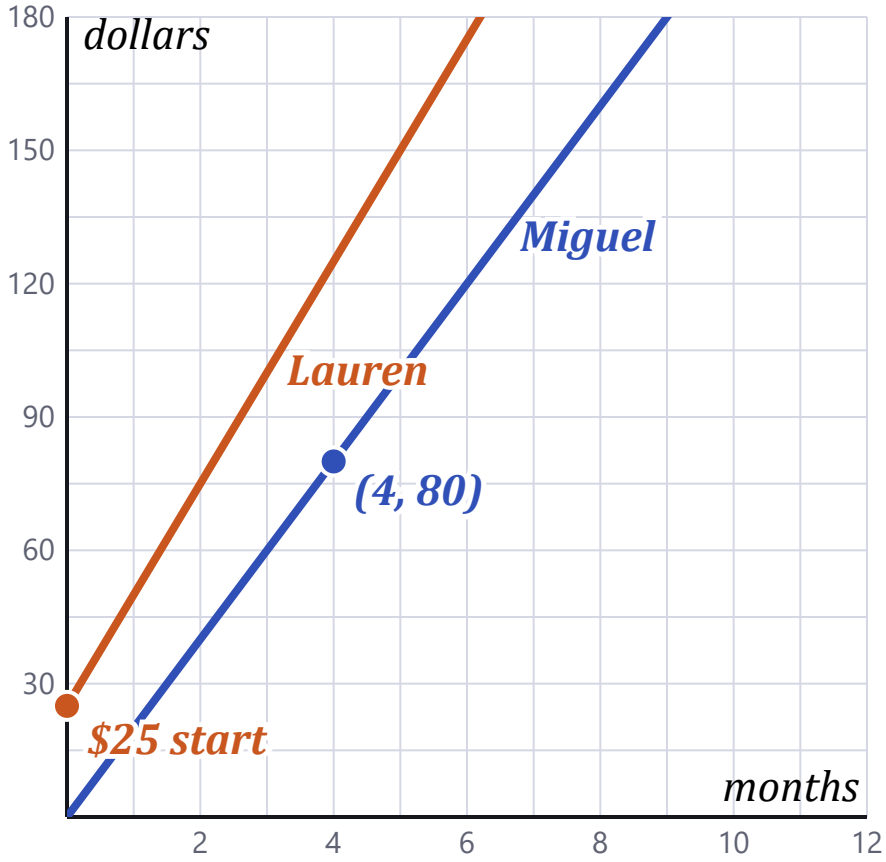
Compare the starts

Both y-intercepts are 0: 0 tons pays $0.

$f(x) = 3x + 10$ vs. a table with $(0, 4)$, $(1, 9)$: the table grows faster (slope 5 > 3), but f starts higher (10 > 4).

Units and Scales

slope unit = output per input



Saving money

Miguel

$0 to start
saves **$20 per month**

Lauren

$25 to start
saves **$25 per month**

Units tell the meaning

Slope: dollars **per** month.
***y*-intercept:** dollars at 0 months.

Choose good scales

Label each axis with the quantity and units. Pick tick sizes so all the data fits (here 1 month and $15).

> Taxi: $2.50 per mile plus $3. Slope 2.50 dollars per mile; *y*-intercept $3.